

Brad Hannah

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AWS AND AZURE CLOUD ARCHITECT | SOLUTION ARCHITECT | SOFTWARE ENGINEER

→ Empowering Enterprises to Build Secure, Scalable Solutions that Drive Business Results and Achieve Digital Transformation ←

AWS Certified Professional with five active AWS certifications, including AWS Professional Architect and previous Azure Architect certifications. Leverage 20 years of expertise including deep expertise across all corners of the project management and execution realm as an executor (analyst, architect, engineer), planner (PM, technology planner), and resource manager.

- Skilled in managing people and projects to build high-performing, results-driven teams, leveraging superb communication skills and an innate ability to collaborate with cross-functional teams. Demonstrated finesse for working closely with product managers to shape features. Passionate about moving the team's mission forward by adopting more modern technologies to reduce operational expenses and respected for superior planning and execution.
- Sought after to optimize and improve existing infrastructure, architecture, and processes to use cloud platforms. Owned all aspects of implementation for features from design to development, deployment to production, to support and maintenance, and consistently designed and built systems that are scalable, highly available, and easily maintainable.
- Demonstrated hands-on programming and development experience with AWS APIs and technologies (EC2, IAM, Route 53, Auto Scaling, Elastic Load Balancing, Elastic Container Service, S3, EBS, VPC, Security Groups, EKS), Immutable Infrastructure and Infrastructure as Code patterns and technologies (Terraform, CDK, CloudFormation, Docker, Ansible, Packer, Vagrant), commercial software development in an Object-Oriented and functional languages (e.g. Python, C#, Go, TypeScript), scripting such as Linux Shell, and open-source technologies such as Nginx, Postgres, Django, Git, Jest, Tomcat, Linux.

CORE EXPERTISE

Cloud-Native Architecture | Systems Architecture | Software Engineering | Technical Systems Engineering | Amazon Web Services (AWS) | Microsoft Azure | Technology Consulting | Multi-Cloud Architecture | Highly-Available Architecture | Python | Go | Distributed Systems | SQL Databases | Object-Oriented Programming (OOP) | Expert Troubleshooting | Technical Leadership | Project Leadership | Requirements Analysis | Process Automation | CI/CD | Observability and Monitoring | Cloud Cost Optimization | Technology Procurement Proposal Writing | Cloud Security and Compliance

PROFESSIONAL EXPERIENCE

AMAZON WEB SERVICES (AWS) – PROFESSIONAL SERVICES/WORLDWIDE PUBLIC SERVICES

2022-Present

Cloud Infrastructure Architect / Technical Consultant

Designed and built innovative solutions that exceeded customer expectations. Oversaw advanced AWS networking and firewall configurations, developing scalable cloud-native solutions and creating tailored Infrastructure as Code (IaC) modules. Effectively managed multiple projects concurrently, interpreting complex business requirements and delivering high-quality solutions on time and within budget.

- Engineered a high-performance Fortinet health monitoring platform for a Canadian government agency, processing millions of logs per hour and providing insights on firewall health and metrics. Automated deployment with CDK and utilized AWS services to ensure scalability and reliability.
- Developed backend APIs in TypeScript with Express.js and Postgres. Optimized and refactored the testing framework (Jest) to improve test writing efficiency and code reliability.
- Implemented a log shipping pipeline that transforms millions of logs per minute ingested from various AWS services into Azure Log Analytics for enhanced SIEM insights.
- Created opinionated Terraform modules for deploying AWS resources, including API Gateway and AWS Batch, to enforce infrastructure consistency for greenfield clients.
- Collaborated with clients to design outbound firewall rules with TLS filtering in Suricata for SAP environments.
- Led the implementation of an Amazon WorkSpaces VDI deployment for a Fortune 500 insurance company, supporting seven Active Directory domains for around 3,000 clients. Developed an AWS cost optimizer using Terraform, Python, TypeScript, and Step Functions to reduce monthly expenses.
- Assisted with the development of a generative AI solution to identify opportunities, providing summaries and analyses, using scraping and processing Lambda functions built in GoLang (Go).
- Built BI dashboards and utilized Athena and Amazon QuickSight to replace legacy Oracle reports.
- Created a custom CloudFormation module for deploying unsupported networking resources.

➤ Left five major clients with maintainable, bespoke solutions that they can maintain and benefit from for years.

QUEEN'S UNIVERSITY

2008-2022

Senior Cloud Architect/Engineer (2019-2022)

Spearheaded and built the University's cloud initiative. Oversaw the migration of many of the University's largest and most complex systems to the Azure cloud, emphasizing DevOps-style automation as a key component of success. Focused on self-service and user empowerment, replacing traditional micro-ticketing processes.

- Developed a self-service cloud automation platform with RunDeck, Terraform, BASH, and Python, reducing VM provisioning times from hours to minutes and enhancing resource management through FortiGate and Azure APIs.
 - Established hybrid network connectivity between the campus and Azure using FortiGate firewalls and Azure VPN Gateway, implementing a hub-and-spoke topology with VNet peering and custom routing.
 - Managed Azure DevOps pipelines for safe daytime deployments of key changes (e.g., web proxy and SSO) on AKS with NGINX/Apache.
 - Created a web app for tracking regulatory controls with Azure Cloud, built on Python, Django, HTML, and Celery. It generated live reports for NIST 800-53, saving time on control audits, saving days or weeks of effort per audit.
 - Improved virtual machine provisioning from over two hours to under 15 minutes, enabling same-day availability and enhancing IT responsiveness and customer satisfaction.
- Designed, developed, and executed a comprehensive migration from on-premises servers to the cloud, significantly reducing the on-premises server inventory as workloads transitioned. Maintained strict control over costs while delivering an exceptional deployment experience for distributed IT staff, resulting in a fully operational and production-ready cloud environment.
- Queen's University is now recognized as one of the leading Canadian higher education institutions in public cloud maturity.

Technology Planner (2017-2018)

Served as a Technology Planner, responsible for creating short- and long-term technology strategies. Conducted thorough analyses of financial, organizational, and customer impacts to guide significant technology decisions.

- Established a greatly simplified server compute environment, improved availability, resiliency, and security across the campus network, and refreshed operational processes that resulted in the improved application, reduced toil, and boosted outstanding accountability.

Project Manager (2016-2017)

Led a one-year secondment that involved managing the Request for Proposals (RFP) to replace an existing PBX-based telephony system with a modern Voice over IP (VOIP) solution. Oversaw RFPs for updating the data center's switching and firewall platforms. Facilitated the transition from traditional PBX telephony to an advanced VOIP system. Successfully managed three major RFPs for networking, firewall, and VOIP replacements, resulting in two successful contracts for the network and firewall upgrades.

- Attained successful RFPs and delivered modernized network and network security equipment.
- Acquired a deeper understanding of costs and efforts required for VOIP modernization for the Queen's campus.

Manager, Systems and Storage (2012-2016)

Managed a team of five Windows Server, Linux, and Solaris admins responsible for all central server and shared storage infrastructure across two primary data centers. Oversaw installation, configuration, and maintenance of multiple operating systems, including Solaris, Windows Server, Red Hat Linux, and Oracle Enterprise Linux.

- Devised a path forward and away from operations-heavy, insecure Solaris systems towards virtualized, more secure, and more carefully controlled Linux.
- Spearheaded the Storage Area Network auto-tiering solution procurement and implementation to reduce optimization toil, improve performance, and simplify troubleshooting.
- Removed extensive, expensive, and operationally heavy tape backup solutions for simpler deduplicating appliances.
- Established a team with daily and weekly checklists, improving accountability and simplifying day-to-day management.

Senior Systems Analyst, Middleware (2012)

Designed and implemented strategic changes, overseeing technical decisions and managing day-to-day operations and technology planning. Modernized single sign-on to implement cloud-ready authentication and authorization.

- Enhanced legacy directory systems for improved security and maintainability.
- Delivered a new single sign-on system compatible with most major SaaS providers, improving security and end-user experience.

Technical Specialist – Applications, Middleware, UNIX Systems Team (2008-2012)

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Provided direction to a junior team member throughout projects. Offered strategic advice to the applications team on system architecture, troubleshooting, and peer reviews of new ideas. Served on-call and provided emergency support for all UNIX systems and services after hours, often diagnosing unfamiliar software issues under time-sensitive conditions.

- Led the successful implementation of an IP address management system and a highly available DHCP infrastructure, which continues to operate smoothly.
- Worked with Oracle to create a migration and licensing strategy for the Sun-based Middleware stack, reworking the licenses to save complicated future renegotiations, ensuring cost efficiency.
- Ensured a highly available and easily maintainable DHCP/DNS/IPAM solution for all campus network connectivity. A solution that runs to this day (2025).
- Drastically reduced Oracle annual licensing costs due to optimizing software deployments and server sizing.

PRIOR EXPERIENCE

LIMESTONE TECHNOLOGIES, Senior Software Developer

- Stabilized primary product code base through improved coding safety techniques and use of newer and safer languages and frameworks.

QUEEN'S UNIVERSITY, Technical Specialist – Special Projects Team

- Modernized and rewrote entire backend for software product used by nearly all staff, students and faculty members. Shifting from C to PHP and MySQL.

EDUCATION, TRAINING & CERTIFICATIONS

St. Lawrence College - Computer Engineering Technology Diploma

- AWS Certified Solutions Architect Professional (January 2024)
- AWS Certified Developer Associate (June 2023)
- AWS Certified Security Specialty (May 2023)
- AWS Certified Solutions Architect Associate (September 2022)
- AWS Certified Cloud Practitioner (July 2022)
- Canadian Government Secret Level Clearance (2022)
- Azure Solution Architect Certificate (2021) (990099767) (Expired)
- Azure Fundamentals Certificate (2020) (990099767)
- Certificate in Enterprise Architecture - University of Toronto (2017)

TECHNICAL EXPERTISE

Technologies (Architectural Experience): Amazon AWS (see below), Microsoft Azure (see below), Microsoft Active Directory, Microsoft Azure Active Directory, TLS/SSL, SAML, Puppet, Ansible, FortiGate firewalls, Hyperconverged Systems, Nutanix, Storage Area Networks, Oracle Peoplesoft, Barracuda Spam Firewall, Office 365, Microsoft IIS, MSSQL, Packer, Checkov

Technologies (First Hand Experience): Fortinet firewalls, Terraform, RunDeck, Azure DevOps, GitHub, GitHub Actions, F5 LTM Load Balancers, Docker, Docker-Compose, Apache web server, NGINX, Postgres, VMware VCenter, VMware ESXi, SVN, Bind DNS, ISC DHCP, Tomcat, Shibboleth Single Sign-on

AWS Technologies (First Hand Experience): Lambda (Python, JavaScript and Go), S3, CloudFront, RDS, Step Functions, SQS, SNS, CloudWatch, CloudTrail, EventBridge, ECS, Fargate, ECR, DynamoDB, EC2, Amazon Workspaces, Managed Active Directory, IAM, API Gateway, Cognito, CodeGuru, CodePipeline, CodeCommit, CodeBuild, Quicksight, Athena, SES, KMS, Private CA, Route 53, Network Firewall, Transit Gateway, VPC, X-Ray, Parameter Store, Secrets Manager, Elastic Load Balancer

Azure Technologies (First Hand Experience): FrontDoor, Application Gateway, Virtual Machines, Log Analytics EventHub, Storage Accounts, Managed Disk, VPN Gateway, Hub and Spoke VNets, Route Tables, Policy, Management Groups, SQL Server, IaaS SQL Server, MySQL as a Service, Azure Kubernetes Service, Load Balancer, App Services/Web Apps, Azure Virtual Desktop, Function Apps, Traffic Manager, Web Application Firewall, Azure Automation, Migrate, Recovery Services Vaults, Network Virtual Appliance, Azure Kubernetes Service

Relevant Languages: Python, GoLang (Go), TypeScript, C#, BASH, HTML, PHP, Powershell, C++, SQL

Frameworks and Libraries: Django, Celery, Gin, HTMX, Express.js, OpenAPI, Gorm, TView, Jest, NUnit, PyTest, Cobra, Viper, TView, Azure SDK, Azure Graph SDK, boto3, .Net Framework (Windows Forms)

Infrastructure As Code: Terraform, CDK, CloudFormation, Ansible

Operating Systems: Ubuntu Linux, CentOS/RedHat Linux, Solaris 8/9/10, Windows Server, MacOS

